

In a hole and shaft assembly of 30 mm nominal size, the tolerances for hole and shaft are as specified below :

$$\text{Hole : } 30^{+0.02}_{-0.000} \text{ mm} \quad \text{Shaft : } 30^{-0.040}_{-0.070} \text{ mm}$$

Determine:

- (i) Maximum and minimum clearance obtainable
- (ii) Allowance
- (iii) Hole and Shaft tolerance
- (iv) MML shaft and hole
- (v) The type of fit.

(iii) Hole : $30^{+0.02}_{-0.00} \text{ mm}$

Upper limit = $30 + 0.02 \text{ mm} = 30.02 \text{ mm}$

Lower limit = $30 - 0.00 \text{ mm} = 30.00 \text{ mm} = \text{MML}_h$

Tolerance = $30.02 - 30.00 = \underline{0.02 \text{ mm}}$

Shaft : $30^{-0.040}_{-0.070} \text{ mm}$

Upper limit : $30 - 0.040 = 29.96 \text{ mm} = \text{MML}_s$

Lower limit : $30 - 0.070 = 29.93 \text{ mm}$

Tolerance = $29.96 - 29.93 = \underline{0.03 \text{ mm}}$

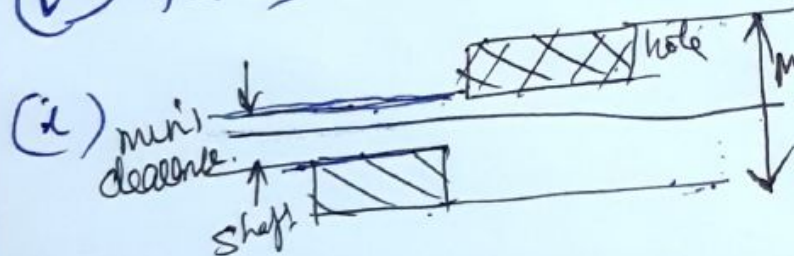
(iv) Allowance = Diff b/w Max. limit of hole & shaft.

= $\text{MML}_h - \text{MML}_s = 30.000 - 29.96 \text{ mm}$

= $\underline{0.04 \text{ mm}}$

$$\begin{array}{r} 30.02 \\ 29.93 \\ \hline 0.09 \end{array}$$

(v) Fit $\Rightarrow$ clearance $\checkmark$



max clearance = $30.02 \text{ mm} - 29.93$
= $\underline{0.09 \text{ mm}}$

min clearance = $30.00 - 29.96 \text{ mm}$
= $\underline{0.04 \text{ mm}}$